

Ideation Phase

Brainstorm & Idea Prioritization

Date	01 OCTOBER 2026
Team ID	SWTID-2026-7129
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization :

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Team gathered and set the goal: brainstorm ways to import spreadsheet data into ServiceNow accurately using Import Sets and Transform Maps. Key rules followed: stay on topic, defer judgment, go for volume, encourage wild ideas, listen to others.

PROBLEM

How might we import spreadsheet data into ServiceNow quickly and accurately, without duplicates or field-mapping errors, using Transform Maps?

Step-2: Brainstorm, Idea Listing and Grouping

Brainstorm — ideas written by each team member:

Team member	Ideas
Aruna V	● Provide a downloadable, pre-formatted Excel template matching the target table ● Validate column headers before upload
Nithya Latchumi R	● Set coalesce on a unique field (e.g., email / serial no.) to avoid duplicates ● Add Transform Map field mapping for every column

Tejaswini J R	● Use onBefore transform script to clean and format data ● Convert reference-field values (e.g., name to sys_id) automatically
Dharshini G P	● Test in a sub-production instance before production ● Use Transform History to log errors per row ● Email notification when an import completes or fails
Kanishka S	● Schedule recurring imports for regular data loads ● Dashboard of import success vs. error rows ● Document a step-by-step guide/checklist for admins

Group ideas — similar ideas clustered under themes:

Cluster	Theme
A. Data preparation & validation	Standardise the spreadsheet before it ever reaches ServiceNow
B. Mapping & transformation logic	Map fields correctly and clean data with transform scripts
C. Duplicate & error handling	Coalesce, ignore/skip rules, and visible error tracking
D. Automation & monitoring	Reuse maps, schedule imports, and notify admins

Step-3: Idea Prioritization

Ideas rated 1–5 on Importance (positive impact if done without cost/difficulty) and Feasibility (cost, time, effort, complexity).

Idea	Importance	Feasibility	Priority
Configure Transform Map with field mappings	5	5	Do first
Set coalesce field to prevent duplicates	5	4	Do first
Standard Excel template + header validation	4	5	Do first

onBefore script for data cleaning / reference lookup	4	3	Plan next
Transform History error review	4	5	Do first
Test import in sub-production instance	5	5	Do first
Email notification on import completion/failure	3	3	Later
Scheduled recurring imports	3	2	Later
Import dashboard (success vs. error)	3	2	Later

Outcome: the core solution focuses on the high-importance, high-feasibility ideas — Transform Map field mapping, coalesce, a standard template, testing in sub-production and error review through Transform History.

Ideation Phase Empathize & Discover

Date	1 OCTOBER 2026
Team ID	SWTID-2026-7129
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Empathy Map Canvas:

Persona: a Data Administrator / HR Operations user who must migrate bulk employee data received in Excel format into ServiceNow accurately, without duplicates.

Think & Feel ● Will this bulk upload create duplicate or wrong employee records? ● Worried a single bad mapping will corrupt data that HR and managers depend on ● Wants onboarding data in the system quickly and accurately, without retyping ● Feels pressure to keep employee data clean, reliable and audit-ready		
Hear ● HR: "We have 200 new joiners in an Excel sheet, please load them today" ● Manager: "Why are some employees missing or listed twice?" ● Peers: "Import Sets stage the data first, then Transform Maps load it" ● "Remember to coalesce on Employee ID"	Data Administrator / HR Operations User Responsible for loading employee data from spreadsheets into ServiceNow	See ● Employee data arriving as Excel / Google Sheets files ● Inconsistent columns, edited names and changed emails across versions ● Import Set staging table, Transform Map and field mappings ● Import logs showing inserts, updates and ignored rows
Say & Do ● "I need to load this spreadsheet without creating duplicates" ● Prepares the sheet, loads it through Load Data and maps fields with Auto Map / Mapping Assist		

● Runs the transform and checks the result in the Transform History ● Re-imports updated sheets to confirm records are updated, not duplicated ● Builds reports and a dashboard to share employee insights		
Pain ● Manual data entry is slow and error-prone ● Duplicate records when the same file is imported again ● Field mismatches between spreadsheet columns and table fields ● No quick visibility into department / location distribution	Gain ● Fast, accurate bulk import with no manual entry ● Coalesce on Employee ID updates existing records and blocks duplicates ● Clean, consistent employee table that can be trusted ● Dashboard with department, location and list views in one place	

Ideation Phase

Define the Problem Statements

Date	1 October 2026
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Maximum Marks	2 Marks

Customer Problem Statement:

Problem statements written from the customer's point of view for the project "Import Data using Transform Maps (Spreadsheet)".

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	a data administrator at a company that receives bulk employee data in Excel	import employee records from a spreadsheet into ServiceNow quickly and accurately	manual entry is slow and fields in the sheet don't match the target table	there is no automated mapping between the spreadsheet columns and the ServiceNow table	frustrated and worried about data errors
PS-2	an HR operations user who re-imports updated employee sheets	update existing employee details without creating duplicate records	the same employee gets inserted again every time the file is re-imported	records are not matched on a unique key such as Employee ID (no Coalesce)	anxious and unable to trust the employee data

PS-3	an HR manager or administrator who reviews employee data	see employee distribution by department and location at a glance	the data sits in a plain table and needs manual counting and analysis	no reports or dashboard are built on the imported data	overwhelmed and unsure of decisions
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Example (reading PS-1 as one sentence):

I am a data administrator at a company that receives bulk employee data in Excel, and I'm trying to import employee records from a spreadsheet into ServiceNow quickly and accurately, but manual entry is slow and fields in the sheet don't match the target table, because there is no automated mapping between the spreadsheet columns and the ServiceNow table, which makes me feel frustrated and worried about data errors.

SCENARIO	CUSTOMER JOURNEY MAP				
	Import Data using Transform Maps (Spreadsheet)				
	User: ServiceNow Administrator Project: bulk employee data import from Excel into ServiceNow				
Importing employee data from a spreadsheet into ServiceNow using Import Sets and Transform Maps	Entice	Enter	Engage	Exit	Extend
	Howdoessomeone initially become aware of this process?	Whatdopeople experience as they begin the process?	In thecore moments in the process, what happens?	Whatdopeople typically experience as the process finishes?	Whathappens after the experience is over?
Steps	Receive employee data – Bulk employee data arrives from the HR / data provider in Excel format.	Prepare the sheet – Create a Google Spreadsheet with sample data (Employee ID, Name, Email, Department, Location).	Load Data – Application Navigator > Load Data; create import table Employee Import (u_employee_import) and upload the file.	Open Employee Test – Openthe table target table from the Application Navigator. Verify the records – Check that all imported records are created in the table.	Enable Coalesce – Set Coalesce to true on a field map so duplicates are updated, not inserted.
What does the person typically experience?	Choose the import method – The admin decides to use Import Sets and Transform Maps to migrate the data efficiently and accurately.	Download as .xlsx – Download the sheet as Sample Spreadsheet.xlsx. Create the target table – Tables > Create New: Employee Test (u_employee_test). Add fields – Add Employee ID, Employee Name, Email, Department, Location (String) using Form Layout.	Create Transform Map – Click Create Transform Map: Sample Spreadsheet Import (source Employee Import, target Employee Test). Map the fields – Use Auto Map Matching Fields and Mapping Assist, then Save. Run the Transform – Click Transform and wait for the success message.	Arrange the columns – Use Personalize List Columns and click OK.	Import updated Excel data – Load the new file into the existing Employee Import table and run the transform again. Check Transform History – Seehow many rows were Inserted and how many were Updated.
Interactions	People: HR team / data provider who sends the employee data Things: Excel / Google Spreadsheet Places: Email or shared drive	Things: Google Sheets, Sample Spreadsheet.xlsx Places: ServiceNow > Tables > Create New Places: Form Layout (Form Context Menu > Configure)	Places: Load Data page in Application Navigator Things: Import set table, Transform Map form, Mapping Assist Things: Transform button and	Places: Employee Test table list view Things: Personalize List Columns dialog	Places: Transform Map > Field Maps related list Places: Load Data (existing table), Transform History Things: updated Excel file
What interactions do they have at each step along the way?					

			success message		
Goals & motivations At each step, what is a person's primary goal? ("Help me...")	- Help me get the employee data into ServiceNow without typing every record - Help me pick the right import approach	- Help me organise the data with the right columns - Help me create a table that matches my data	- Help me load the file without errors - Help me match every source field to the right target field - Help me be sure the transform actually ran	- Help me confirm all records were imported correctly - Help me see the data in a readable order	- Help me avoid duplicate records when data is imported again - Help me update existing records safely - Help me see what was inserted and what was updated
Positive moments What steps does a typical person find enjoyable, productive or delightful?	Bulk data can be moved in one go instead of record by record	- Table name is auto-populated from the label - New fields show up on the Employee Test form after saving	- Auto Map Matching Fields fills in matching fields quickly - Success messages appear after Submit and Transform	- All records appear in the Employee Test table - Final result: data successfully imported	- Matching IDs are updated and no duplicates are created - Transform History clearly shows Inserted and Updated counts
Negative moments What steps does a typical person find frustrating, confusing or time-consuming?	- Beginners may not know where to start - Data may not be in the layout ServiceNow expects	- Finding Form Layout takes several clicks - Each of the five fields has to be added one by one	- Wrong field mapping would give wrong data - Many screens to follow; easy to miss the Create Transform Map step	Columns may not appear in the expected order and need manual rearranging	- Without Coalesce, importing again creates duplicate records - Choosing the right field for Coalesce needs care
Areas of opportunity How might we make each step better? What ideas do we have?	- Provide a sample spreadsheet template with the required columns - Share a short guide of the whole import process	- Give a checklist of fields to create - Create fields automatically from the spreadsheet columns	- Highlight the Create Transform Map button after loading data - Add a quick check for unmapped fields before transforming	- Save a default list layout for imported records - Add a filtered view showing only newly imported records	- Enable Coalesce on Employee ID from the start - Reuse the same import table and transform map for future files

Project Design Phase-II

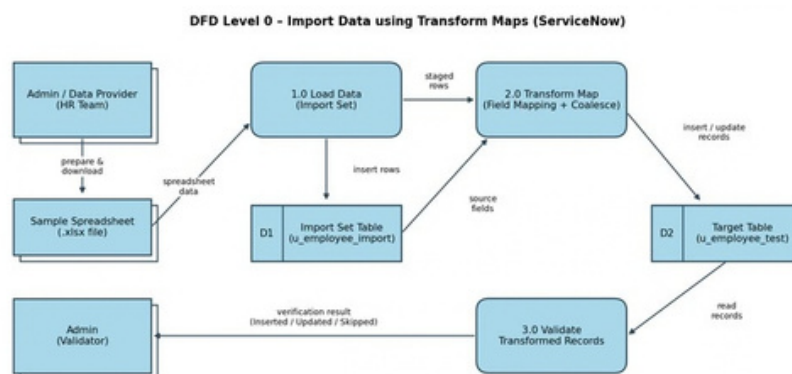
Data Flow Diagram & User Stories

Date	01 OCTOBER 2026
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Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: DFD Level 0 (Project – Import Data using Transform Maps)



Flow:

1. Data provider prepares the employee data in a Google Spreadsheet and downloads it as Sample Spreadsheet.xlsx.
2. Administrator opens Load Data and uploads the file into the import set table Employee Import (u_employee_import), which acts as the staging area.
3. The Transform Map (Sample Spreadsheet Import) maps the source fields to the target table fields, with Coalesce used to avoid duplicates.
4. On running the Transform, records are inserted or updated in the target table Employee Test (u_employee_test).
5. Administrator validates the records in the Employee Test table and checks Transform History (Inserted / Updated / Skipped).

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Data Provider (HR Team)	Prepare the Sheets	USN-1	As a data provider, I can create a Google Spreadsheet with sample employee data (Employee ID, Employee Name, Email, Department, Location).	The spreadsheet contains all sample employee records with the correct columns.	High	Sprint-1
		USN-2	As a data provider, I can download the spreadsheet in .xlsx format.	Sample Spreadsheet.xlsx is downloaded and ready to upload.	High	Sprint-1
Administrator	Create Custom Table	USN-3	As an administrator, I can create a new custom table with the label Employee Test (u_employee_test).	The Employee Test table is created and saved successfully.	High	Sprint-1
		USN-4	As an administrator, I can add the fields Employee ID, Employee Name, Email, Department and Location (type String) using Form Layout.	All five fields are displayed on the Employee Test form after saving.	High	Sprint-1
	Import Set Table	USN-5	As an administrator, I can use Load Data to upload the spreadsheet into a new import set table, Employee Import (u_employee_import).	A success message is shown and the Create Transform Map option is available.	High	Sprint-2
	Transform Map	USN-6	As an administrator, I can create a transform map named Sample Spreadsheet Import with Employee Import as the source table and Employee Test as the target table.	The transform map is saved with the correct source and target tables.	High	Sprint-2

		USN-7	As an administrator, I can use Auto Map Matching Fields and Mapping Assist to map source fields to target fields.	All source and target table fields are mapped correctly.	High	Sprint-2
		USN-8	As an administrator, I can run the Transform to move data from the import set table to the target table.	A success message is displayed after the transformation completes.	High	Sprint-2
	Transform Data & Validate	USN-9	As an administrator, I can open the Employee Test table and verify that all imported records are created, arranging columns using Personalize List Columns.	All spreadsheet records appear in the Employee Test table.	Medium	Sprint-3
	Enable Coalesce	USN-10	As an administrator, I can set Coalesce to true on a field map so that duplicate records are updated instead of inserted again.	Re-importing existing employee IDs updates the records and creates no duplicates.	Medium	Sprint-3
	Insert New Data in Excel Format	USN-11	As an administrator, I can load a new Excel file into the existing Employee Import table and run the transform map again.	The import set state shows Completed and Success.	Medium	Sprint-3
		USN-12	As an administrator, I can check Transform History to see which rows were inserted and which were updated.	Transform History shows Medium 2 rows inserted (new IDs) and 2 rows updated (changed name/email).		Sprint-3

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	01 OCTOBER 2026
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Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Prepare Source Data (Spreadsheet)	Create a Google Spreadsheet with sample employee data (Employee ID, Employee Name, Email, Department, Location) Download the spreadsheet as Sample Spreadsheet.xlsx
FR-2	Create Custom Target Table	Create the Employee Test table (u_employee_test) using Tables > Create New Add fields Employee ID, Employee Name, Email, Department and Location (type String) using Form Layout
FR-3	Load Data into Import Set Table	Upload the .xlsx file using Load Data (Sheet number 1, Header row 1) Create the import set table Employee Import (u_employee_import) as the staging area
FR-4	Transform Map Configuration	Create the transform map Sample Spreadsheet Import (source: Employee Import, target: Employee Test) Map source and target fields using Auto Map Matching Fields and Mapping Assist Run the Transform to create records in the target table
FR-5	Transform Data & Validation	Open the Employee Test table and verify that all records are imported Arrange the columns using Personalize

		List Columns Check Transform History for Inserted / Updated / Skipped records
FR-6	Avoid Duplicate Records (Coalesce)	Set Coalesce to true on a field map so existing records are updated instead of duplicated Re-import new Excel data into the existing Employee Import table and run the transform again

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The import is done through simple UI steps (Load Data, Auto Map Matching Fields, Mapping Assist) that need no coding, so beginners can follow it easily.
NFR-2	Security	Only authorised users (administrators) can create tables, import set tables and transform maps, and the data stays within the ServiceNow instance.
NFR-3	Reliability	Success messages are shown at each step and Transform History records the result. Coalesce keeps data consistent by preventing duplicate records on repeated imports.
NFR-4	Performance	Bulk employee data is first loaded into the import set staging table and then moved to the target table in a single Transform run.
NFR-5	Availability	The import runs on the ServiceNow platform, and Load Data can be repeated at any time using the existing Employee Import table.
NFR-6	Scalability	The same import set table and transform map can be reused for new files, and more rows or fields can be added by extending the table fields and field maps.

PROJECT DESIGN PHASE – II

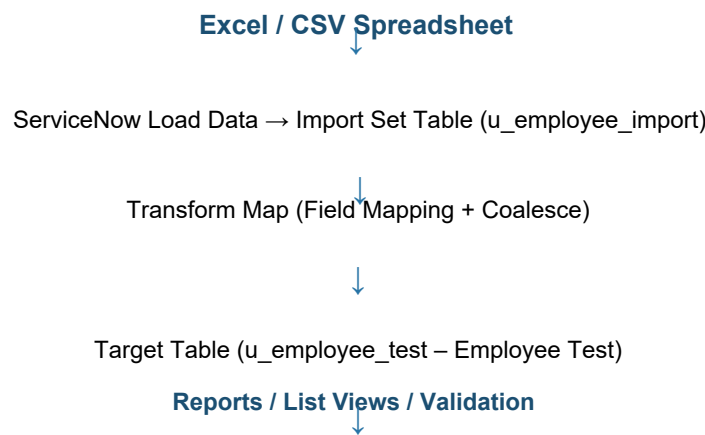
Technology Stack

(Architecture & Stack)

Project Name	Import Data using Transform Maps (Spreadsheet)
Platform	ServiceNow (SaaS / Cloud Platform)
Document Type	Technology Stack – Architecture & Components

1. High-Level Architecture Flow

The data import process follows a clear, sequential pipeline native to the ServiceNow platform:



2. Components & Technologies

S.No	Component	Description	Technology / Tool
1	User Interface	Upload spreadsheet, configure maps, view results	ServiceNow Web UI
2	Data Loading	Loads spreadsheet into a temporary staging table	ServiceNow Import Sets (Load Data)
3	Field Mapping & Logic	Maps source fields to target fields; handles coalesce	ServiceNow Transform Maps

4	Transformation Engine	Executes mapping, insert / update / ignore logic	ServiceNow Transform Engine
5	Target Data Store	Stores the final employee records	Custom Table: u_employee_test
6	Staging Data Store	Temporary holding area for imported data	Import Set Table: u_employee_import
7	Database	Persistent storage for all tables and records	ServiceNow Database
8	File Format	Source of bulk employee data	Microsoft Excel (.xlsx) / CSV
9	Reporting	Visualize imported data (Department, Location)	ServiceNow Reports (Pie / Bar)
10	Security & Access	Authentication, roles, and data protection	ServiceNow ACLs & Access Controls
11	Infrastructure	Hosts the entire application	ServiceNow Cloud Platform (SaaS)
12	External APIs	Not required for this project	Not Applicable
13	Machine Learning	Not required for this project	Not Applicable

3. Application Characteristics

S.No	Characteristic	Description	Technology
1	Platform	Fully managed low-code / no-code platform	ServiceNow Platform
2	Open-Source	No external open-source framework is	ServiceNow Native

	Frameworks	required	
3	Security	Role-based access, ACLs, and audit logs protect data	ServiceNow Security Model
4	Scalability	Cloud platform supports scalable bulk data processing	ServiceNow Cloud
5	Availability	High availability through ServiceNow SaaS environment	ServiceNow Cloud Infrastructure
6	Performance	Efficient transform of spreadsheet records with coalesce	Import Sets & Transform Maps
7	Data Integrity	Coalesce on unique key prevents duplicate records	Transform Map Coalesce
8	Extensibility	Easy to add fields, scripts, or additional reports later	ServiceNow Tables & Forms

4. Recommended Technology Choices (Summary)

Layer	Choice	Reason
Platform	ServiceNow	Project is 100% ServiceNow native
User Interface	ServiceNow Platform UI	No external frontend required
Data Import	Import Sets + Transform Maps	Standard & recommended ServiceNow approach for spreadsheet migration
Database	ServiceNow Database	Managed, integrated, and secure
File Source	Microsoft Excel (.xlsx)	Easy for business users to prepare and share

Reporting	Native ServiceNow Reports	Pie and Bar charts on Department & Location
Hosting	ServiceNow Cloud (SaaS)	Zero infrastructure management overhead

5. Key ServiceNow Artifacts Used

- Custom TargetTable: u_employee_test (Employee Test)
- Import Set Table: u_employee_import (Employee Import)
- Transform Map: Sample Spreadsheet Import
- Coalesce Field: u_employee_id (Employee ID) – prevents duplicate records
- Reports: Employees by Department (Pie Chart), Employees by Location (Bar Chart)

6. Conclusion

This project is a pure ServiceNow low-code solution. No external frameworks, custom backend services, machine learning models, or third-party APIs are required.

All functionality is delivered using native ServiceNow capabilities: Import Sets, Transform Maps, custom tables, coalesce logic, and Reports. This approach ensures rapid development, high reliability, built-in security, and easy maintainability.

— End of Technology Stack Document —

Project Design Phase

Problem - Solution Fit

Project: Import Data using Transform Maps (Spreadsheet) | Team ID: SWTID-2026-7129

Purpose / Vision: Automate bulk employee data import from spreadsheets into ServiceNow so records are accurate, duplicate-free and visible through reports and dashboards.

1. CUSTOMER SEGMENT(S) ● Data administrators and HR operations users in organisations that use ServiceNow ● Receive bulk employee data as Excel / Google Sheets files ● Beginner to intermediate ServiceNow users	6. CUSTOMER CONSTRAINTS ● Data arrives in spreadsheet format, not directly in the platform ● Limited ServiceNow knowledge of Import Sets and Transform Maps ● Time pressure to load large files, and no room for duplicate or wrong records ● Needs admin access (e.g. ACL / admin overrides) to build tables, reports and dashboards	5. AVAILABLE SOLUTIONS ● Manual entry record by record: simple but slow and error-prone ● Direct table import from the list view: quick but no field mapping control and no duplicate check ● Custom scripts or integrations: powerful but need coding skills and maintenance
2. JOBS-TO-BE-DONE / PROBLEMS ● Import employee data from a spreadsheet into ServiceNow quickly and accurately ● Map spreadsheet columns to the correct target table fields ● Avoid duplicate records when the same data is imported again ● Update existing employee records instead of inserting new ones ● See employee data as reports (by department and location) on a dashboard	9. PROBLEM ROOT CAUSE ● Employee data is created and maintained outside ServiceNow, in spreadsheets ● Spreadsheet columns do not match table fields, so mapping is needed ● Without a unique key (Coalesce on Employee ID) every import inserts new rows ● Imported data has no reports or dashboards built on it	7. BEHAVIOUR ● Prepares a spreadsheet and downloads it as .xlsx ● Loads it through Load Data into an Import Set staging table ● Creates a Transform Map, maps fields and runs the transform ● Checks Transform History for inserts, updates and ignored rows ● Re-imports updated sheets and creates reports and a dashboard
3. TRIGGERS ● A new batch of employees joins and their details arrive in Excel ● HR sends an updated sheet with changed names or emails ● Duplicate or outdated records are noticed in the employee table ● Management asks for employee headcount by department or location	10. YOUR SOLUTION ● Load the spreadsheet into an Import Set table (u_employee_import) ● Transform Map with Auto Map / Mapping Assist loads data into the Employee Test table (u_employee_test) ● Coalesce on Employee ID: existing records are updated, duplicates are ignored ● 3 reports (Employees by Department, Employees by Location, Employee List) on the Employee Analytics Dashboard	8. CHANNELS of BEHAVIOUR 8.1 ONLINE ● Google Sheets to prepare and download the .xlsx file ● ServiceNow Load Data, Transform Map and Transform History ● Reports and dashboard in ServiceNow 8.2 OFFLINE ● Team and HR discussions on which data must be loaded ● Emailed or shared Excel files from HR ● Verifying records against the original sheet
4. EMOTIONS: BEFORE / AFTER ● BEFORE: stressed, worried about errors and duplicates, frustrated by repetitive manual entry ● AFTER: confident and in control, relieved that data is accurate, clean and easy to see on a dashboard		

Project Design Phase Proposed Solution Template

Date	01 OCTOBER 2026
Team ID	SWTID-2026-7129
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	2 Marks

Proposed Solution :

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Organizations frequently receive bulk employee or master data in Excel/CSV format that must be migrated into ServiceNow. Manual data entry is time-consuming, error-prone, and does not scale. Without a structured import mechanism, duplicate records, incorrect field mapping, and data integrity issues arise, leading to poor data quality and operational inefficiency.
2.	Idea / Solution description	The solution uses ServiceNow native capabilities – Import Sets and Transform Maps – to efficiently import structured spreadsheet data into a target table (Employee Test). Key steps: • Load Excel data into an Import Set staging table (u_employee_import) • Create a Transform Map to map source fields (Employee ID, Name, Email, Department, Location) to the target table (u_employee_test) • Enable Coalesce on Employee ID to prevent duplicates and support updates

		• Execute the transform and validate results • Generate reports (Employees by Department, Employees by Location) for insights
3.	Novelty / Uniqueness	• Leverages ServiceNow's low-code platform – no custom backend or external tools required • Coalesce feature automatically handles insert vs update logic, ensuring data integrity on repeated imports • Provides a reusable, configurable pattern that can be extended to any master data import scenario (assets, locations, CI data, etc.) • Combines data import with immediate reporting for end-to-end visibility • Fully auditable through Transform History and Import Logs
4.	Social Impact / Customer Satisfaction	• Significantly reduces manual effort and human error in data migration tasks • Improves data accuracy and consistency across the organization • Enables faster onboarding of employees or bulk data updates, improving operational readiness • Increases user/admin satisfaction by providing a guided, reliable, and repeatable import process • Supports compliance and audit requirements through complete import history and logs
5.	Business Model (Revenue Model)	This is an internal process-improvement solution within the ServiceNow platform. It does not generate direct revenue but delivers strong cost savings and efficiency gains: • Reduced labor cost from eliminating manual data entry • Lower risk of costly data errors and rework • Faster time-to-value for HR, IT, and operations teams • Can be packaged as a reusable best-practice accelerator or consulting offering for ServiceNow implementation partners
6.	Scalability of the Solution	• ServiceNow Import Sets and Transform Maps are designed to handle large volumes of records efficiently • The same Transform Map can be reused for multiple import runs without reconfiguration • Coalesce ensures the solution remains clean even when the same data is imported repeatedly

		• Easily extended to additional fields, tables, or data sources (CSV, XML, etc.) • Runs on ServiceNow Cloud infrastructure which provides inherent horizontal scalability and high availability • Supports scheduled or on-demand imports, making it suitable for both one-time migrations and ongoing data synchronization
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— End of Proposed Solution Template —

Project Design Phase Solution Architecture

Date	15 February 2025
Team ID	
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Solution Architecture:

The solution bridges a common business problem, data stored in spreadsheets that must be entered into ServiceNow by hand, with a ServiceNow-native technology solution: Import Sets and Transform Maps. A spreadsheet is loaded into a staging (import set) table, and a transform map converts each row into a record on the target table with the right field mapping, validation and duplicate handling. The architecture aims to:

- Find the best tech solution: use ServiceNow's built-in Import Set and Transform Map engine, so no external ETL tool or custom integration is needed.
- Describe the structure and behavior of the solution to stakeholders: file upload, staging, transformation, target update, logging.
- Define features, development phases and requirements: data source, import set table, field maps, coalesce, transform scripts, scheduled or manual runs.
- Provide specifications by which the solution is built, managed and delivered: tables, roles, mapping rules and validation checks listed in this document.

Solution Architecture Diagram:

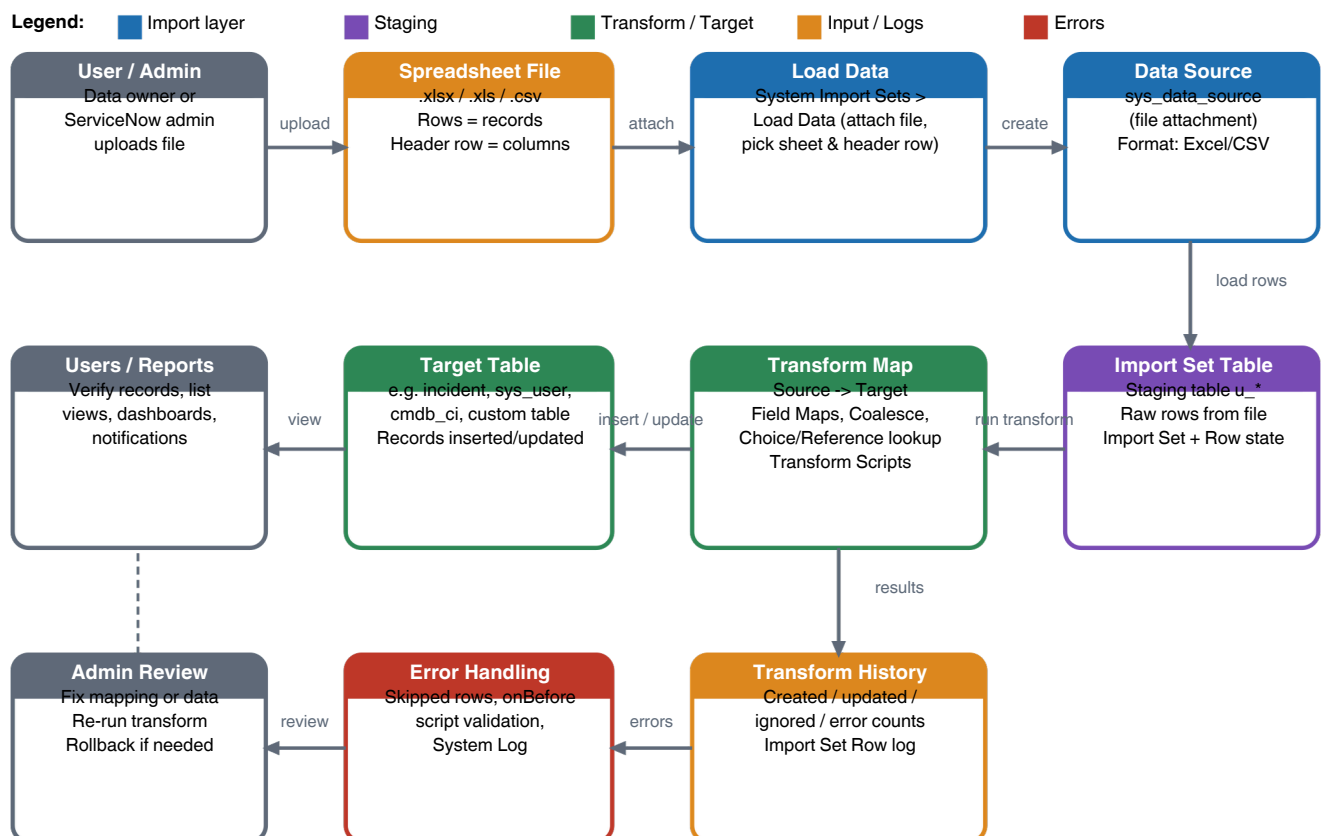


Figure 1: Architecture and data flow for importing spreadsheet data into ServiceNow using Transform Maps

Components and Responsibilities

Component	ServiceNow artifact	Responsibility
User / Admin	Roles: admin, import_admin, import_transformer	Prepares the spreadsheet, runs the import, reviews results.
Spreadsheet file	.xlsx / .xls / .csv	Source data. First row holds column headings; one record per row.
Load Data	System Import Sets > Load Data	Creates the import set table, attaches the file, chooses sheet number and header row.
Data Source	sys_data_source	Stores file type (Excel/CSV), sheet, header row and the attachment.
Import Set Table	Staging table (u_ prefix), extends Import Set Row	Holds raw rows exactly as in the file so the target table is never touched until the transform runs.
Transform Map	sys_transform_map and field maps	Maps source columns to target fields; coalesce field prevents duplicates; choice and reference lookups; onBefore / onAfter scripts for validation.
Target Table	e.g. incident, sys_user, cmdb_ci or a custom table	Receives inserted or updated records.
Transform History / Logs	sys_import_set_run, Import Set Rows, System Log	Shows inserted, updated, ignored and error counts per row for auditing.
Error Handling	Row state = Error / Ignored, transform scripts	Invalid rows are skipped with a reason; admin fixes data or mapping and re-runs.

Data Flow

1. The user prepares the spreadsheet with a header row and clean column values.
2. In Load Data, the user creates a new import set table, attaches the file and selects the sheet and header row.
3. ServiceNow creates a Data Source and loads every row into the staging (import set) table.
4. The user creates a Transform Map from the staging table to the target table and maps fields (auto map plus manual mapping).
5. A coalesce field (for example email or number) is set so existing records are updated instead of duplicated.
6. The user runs the transform. Each row is validated, mapped and inserted or updated in the target table.
7. Transform History shows the counts and any errors. Admin corrects problems and re-runs or rolls back the import set.
8. End users verify the new records through list views, reports or dashboards.

Key Design Considerations

- **Data quality:** staging table separates raw data from live data; onBefore scripts reject blank or invalid rows.
- **Duplicate control:** coalesce on a unique field; "Run business rules" setting chosen per target table.
- **Security:** only users with import roles can load data; ACLs on the target table still apply.
- **Scalability:** large files can be loaded in batches or via a Scheduled Import; Rollback available for bad loads.
- **Maintainability:** reusable transform map for each recurring file format.

Reference: ServiceNow Product Documentation - Import sets, Transform maps and Load Data (docs.servicenow.com).

A Sprint fixed period or duration in which a team works to complete a set of tasks

An Epic is a big task or project that is too large to complete in one sprint. It is broken down into smaller tasks (stories) that can be completed over multiple sprints.

A Story is a small task . It is part of an Epic.

A Story Point is a number that represents how much effort a story takes to complete.
(usually in form of Fibonacci series)

1-Very Easy task

2-Normal task

3-Moderate task

4-Difficult task

Storypoint -4 (1,2,3,4)

Sprint 1

Data Collection (Epic 1)

Gathering Data (USN1) 2

Loading Data(USN2) 1

Data Preparation (Epic 2)

Handling Missing Values(USN3) 3

Creating Fields(USN4)3

Handling Inconsistency in Data(USN5)3

Total Story Point in Sprint 1= 2+1+3+3+3=12

Sprint 2

Data Visualization (Epic 3)

Bar Chart(USN6) 2

Pie Chart(USN7) 2

Line Chart(USN8) 2

Map(USN9) 4

Dashboard (Epic 4)

Developing Dashboard(USN10) 5

Story (Epic 5)

Developing Story(USN11) 5

Total story Point in Sprint 2= 2+2+2+4+5+5=20

Total Story Points

Sprint 1 = 12

Sprint 2 = 20

Velocity= Total Story Points Completed/ Number of Sprints

Total story Points= 12+20=32

No of Sprints= 2

Velocity =32/2

16(Story Points per Sprint)

Your team's velocity is 16 Story Points per Sprint.

Project Planning – Sprint Planning & Estimation

Import Data using Transform Maps (Spreadsheet) – ServiceNow

Team Members: Aruna V, Nithya Latchumi R, Tejaswini J R, Dharshini G P, Kanishka S

1. Planning Terms

- **Sprint:** a fixed period in which the team completes a set of tasks.
- **Epic:** a large piece of work too big for one sprint, broken down into stories.
- **Story:** a small task that is part of an Epic.
- **Story Point:** a number showing the effort a story needs, using the Fibonacci-style scale below.

2. Story Point Scale

Points	Meaning
1	Very easy task
2	Normal task
3	Moderate task
5	Difficult task

3. Sprint 1 – Data Preparation and Import Setup

Story ID	Epic	User Story	Story Points	Assigned To
USN-1	Data Preparation (Epic 1)	Prepare the spreadsheet with source data	2	Aruna V
USN-2	Data Preparation (Epic 1)	Clean and validate spreadsheet columns	3	Nithya Latchumi R
USN-3	Import Setup (Epic 2)	Create target table in ServiceNow	3	Tejaswini J R

USN-4	Import Setup (Epic 2)	Upload spreadsheet to the import set table	2	Dharshini G P
USN-5	Import Setup (Epic 2)	Create the transform map	5	Kanishka S
Total Story Points			15	

4. Sprint 2 – Mapping, Validation and Documentation

Story ID	Epic	User Story	Story Points	Assigned To
USN-6	Field Mapping (Epic 3)	Map source fields to target fields	3	Tejaswini J R
USN-7	Field Mapping (Epic 3)	Configure coalesce to avoid duplicates	2	Aruna V
USN-8	Field Mapping (Epic 3)	Add choice conversion / field scripts	3	Kanishka S
USN-9	Validation (Epic 4)	Run the transform	2	Dharshini G P
USN-10	Validation (Epic 4)	Resolve errors and skipped rows	3	Nithya Latchumi R
USN-11	Validation (Epic 4)	Verify imported records in the target table	2	Aruna V
USN-12	Documentation (Epic 5)	Final documentation and demo	3	Team
Total Story Points			18	

5. Velocity Calculation

Sprint 1 total = $2 + 3 + 3 + 2 + 5 = 15$

Sprint 2 total = $3 + 2 + 3 + 2 + 3 + 2 + 3 = 18$

Total story points = $15 + 18 = 33$

Number of sprints = 2

Velocity = Total story points completed $\div$ Number of sprints = $33 \div 2 = 16.5$

The team's velocity is 16.5 story points per sprint.

Project Development Phase

Import Data using TransformMaps– Functional & Performance Test

Date	01 OCTOBER 2026
Team ID	SWTID-2026-7129
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	

Import Data using Transform Maps – Functional & Performance Test

Project team shall fill the following information in model performance testing.

S.No.	Parameter	Values	Screenshot
1	Import Setup	ServiceNow Import Set receives the spreadsheet and creates import/staging records.	Setup
2	Field Mapping	Source spreadsheet columns map to the intended target ServiceNow fields.	Mapping
3	Transform Execution	Transform Map converts source data and creates/updates target records according to configuration.	Execution
4	Data Validation	Invalid/missing source values are identified and failed rows are visible in import results.	Validation
5	Duplicate Handling	Configured matching/coalesce behavior prevents unintended duplicate records.	Handling
6	Result Verification	Target records contain the expected mapped/transformed values after the import.	Verification

User Acceptance Testing (UAT)

Date	01 OCTOBER 2026
Team ID	SWTID-2026-7129
Project Name	Import Data using Transform
Maximum Marks	Maps(Spreadsheet)

Project Overview:

Project Name: Import Data using Transform Maps(Spreadsheet)

Project Description: Import spreadsheet data into ServiceNow using Import Sets and Transform Maps, including field mapping, transformation, validation, matching, and result verification.

Project Version: 1.0

Testing Period: [24 September 2026 to [30 September 2026]

Testing Scope: Spreadsheet upload/import, source-to-target field mapping, transformation logic, validation/error handling, duplicate/matching behavior, and target-record verification.

[List of Features and Functionalities to be Tested]

[List of User Stories or Requirements to be Tested]

Testing Environment:

URL/Location: [Enter ServiceNow instance/application URL]

Credentials (if required): [Use authorized test credentials; do not include passwords in the document]

Test Cases:

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	Valid spreadsheet import	Upload a valid spreadsheet and run the import	Import rows are created in the import/staging table	[Record actual result]	[Pass/Fail]
TC-002	Field mapping	Run the Transform Map with correctly mapped source columns	Values appear in the intended target fields	[Record actual result]	[Pass/Fail]
TC-003	Transformation	Use source values requiring configured transformation	Target values follow the transformation rule	[Record actual result]	[Pass/Fail]
TC-004	Missing/invalid data	Import rows containing missing or invalid required values	Invalid rows are identified and error details are available	[Record actual result]	[Pass/Fail]

TC-005	Duplicate/matching	Import a row matching an existing record according to the configured matching rule	Record is handled according to the configured coalesce/matching behavior	[Record actual result]	[Pass/Fail]
TC-006	Target verification	Open transformed target records after import	Records contain expected values and no unintended data changes	[Record actual result]	[Pass/Fail]

Bug Tracking:

Bug ID	Bug Description	Steps to reproduce	Severity	Status	Additional feedback
BG-001	[Record any import/mapping defect found during UAT]	[Steps to reproduce]	[Low/Medium/High]	[Open/In Progress/Closed]	[Additional feedback]
...	...	...	...	...	...

Sign-off:

Tester Name: Iniyan

Date: 30 September 2026

Signature: Iniyan

Notes:

- Ensure that all test cases cover both positive and negative scenarios.
- Encourage testers to provide detailed feedback, including any suggestions for improvement.
- Bug tracking should include details such as severity, status, and steps to reproduce.
- Obtain sign-off from both the project manager and product owner before proceeding with deployment.

Acceptance Testing – UAT Execution & Report Submission

Date: 30 September 2026
Team ID: [Enter Team ID]
Project Name: Import Data using Transform Maps (Spreadsheet)
Maximum Marks: 4 Marks

1. Purpose of Document

This document records the user acceptance test coverage and open issues for the ServiceNow spreadsheet import project using Import Sets and Transform Maps.

2. Defect Analysis

Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
By Design	0	0	0	0	0
Duplicate	0	0	0	0	0
External	0	0	0	0	0
Fixed	[Actual]	[Actual]	[Actual]	[Actual]	[Actual]
Not Reproduced	0	0	0	0	0
Skipped	0	0	0	0	0
Won't Fix	0	0	0	0	0
Totals	[Actual]	[Actual]	[Actual]	[Actual]	[Actual]

3. Test Case Analysis

Test Case	Scenario	Expected Result	Actual Result	Status
TC-001	Valid spreadsheet import	Import rows are created in staging/import table.	[Actual]	[Pass/Fail]
TC-002	Field mapping	Values reach intended target fields.	[Actual]	[Pass/Fail]
TC-003	Transformation	Configured transformations are applied.	[Actual]	[Pass/Fail]
TC-004	Invalid/missing data	Errors are identified for correction.	[Actual]	[Pass/Fail]
TC-005	Duplicate/matching	Configured matching behavior is followed.	[Actual]	[Pass/Fail]
TC-006	Target verification	Target records contain expected values.	[Actual]	[Pass/Fail]

4. UAT Sign-off

Tester Name: [Enter name]
Date: [Enter completion date]
Signature: [Enter signature]

Note: Replace [Actual], [Pass/Fail], and sign-off placeholders after executing the project tests. Do not include passwords or secrets.

Project Report

Import Data using Transform Maps(Spreadsheet)

ServiceNow Platform | Import Sets, Transform Maps, Reports & Dashboards

1. INTRODUCTION

1.1 Project Overview

This micro project focuses on importing structured data from an external spreadsheet into the ServiceNow platform using Import Sets and Transform Maps. It simulates a real-world scenario where bulk employee data is received in Excel format and must be migrated into ServiceNow efficiently and accurately.

The spreadsheet data is first loaded into an Import Set table, which serves as a staging area. A Transform Map is then configured to map the source fields to the appropriate target table fields. On executing the transformation, records are created in the target table and validated to ensure data integrity. The project was later extended with Coalesce (to avoid duplicate records), Reports and a Dashboard.

1.2 Purpose

- To understand the end-to-end data import process in ServiceNow: data loading, field mapping, transformation execution and result verification.
- To migrate bulk employee data from an Excel spreadsheet into a custom table (Employee Test) accurately and efficiently.
- To prevent duplicate records on repeated imports by using Coalesce on the Employee ID field.
- To provide visibility into employee data through Reports and a consolidated Dashboard.

2. IDEATION PHASE

2.1 Problem Statement

Organizations frequently receive employee data in spreadsheet format from HR or external systems. Entering this data manually into ServiceNow is slow and error-prone, and importing the same file more than once can create duplicate employee records. There is a need for a reliable, repeatable way to bulk-load spreadsheet data, update existing records instead of duplicating them, and monitor the data visually.

2.2 Empathy Map Canvas

User persona: ServiceNow administrator / HR data coordinator who receives employee data in Excel and is responsible for keeping the employee table accurate.

Quadrant	Description
Says	"I have to load this Excel file into ServiceNow." "Please don't create duplicate employees." "Can I see the department-wise count quickly?"
Thinks	Manual entry will take too long and may introduce mistakes. Repeated uploads could corrupt existing data. I need proof the import worked.
Does	Prepares the sheet, downloads it as .xlsx, loads it into an Import Set, runs the Transform Map, and checks the target table.
Feels	Worried about data accuracy and duplicates; relieved when the transform shows Success; confident when reports and dashboard confirm the data.

2.3 Brainstorming

Different approaches were considered for moving spreadsheet data into ServiceNow:

Idea	Pros / Cons	Decision
Enter each record manually	Simple, but slow and error-prone for bulk data	Rejected
Custom script / integration	Flexible, but needs coding and is too heavy for a beginner-level use case	Rejected
Import Set + Transform Map	No-code, staged, validated, reusable for repeated imports	Selected

Coalesce on Employee ID	Updates existing records and skips duplicates on re-import	Selected
Reports + Dashboard	Gives visual insight into department, location and employee lists	Selected

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Action	Touchpoint	Outcome
Prepare	Create sample employee data in Google Sheets and download as .xlsx	Google Sheets	Sample Spreadsheet.xlsx ready
Set up	Create custom table Employee Test with 5 fields	Tables > Create New	Target table ready
Load	Load spreadsheet into an Import Set table	System Import Sets > Load Data	u_employee_import created
Transform	Create Transform Map, map fields, run Transform	Transform Map / Mapping Assist	Records created in Employee Test
Validate	Verify records; enable Coalesce; re-import changed data	Employee Test list, Transform History	Inserted / updated / ignored counts confirmed

Analyze	Create 3 reports and add them to a dashboard	Reports, Dashboards	Employee Analytics Dashboards
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3.2 Solution Requirement

Functional requirements

- Create a custom table (Employee Test, u_employee_test) with fields: Employee ID, Employee Name, Email, Department and Location (all String).
- Load data from an Excel (.xlsx) file into an Import Set table (Employee Import, u_employee_import).
- Create a Transform Map (Sample Spreadsheet Import) mapping source fields to the target table fields.
- Enable Coalesce so that existing records are updated and duplicates are not inserted.
- Generate reports (Employees by Department, Employees by Location, Employee List Report) and a dashboard.

Non-functional requirements

- Accuracy and data integrity of imported records.
- Repeatability: the same import can be run multiple times without creating duplicates.
- Usability: no-code, configuration-based steps that a beginner can follow.
- Access control: dashboard sharing can be restricted by users, groups and roles.

3.3 Data Flow Diagram

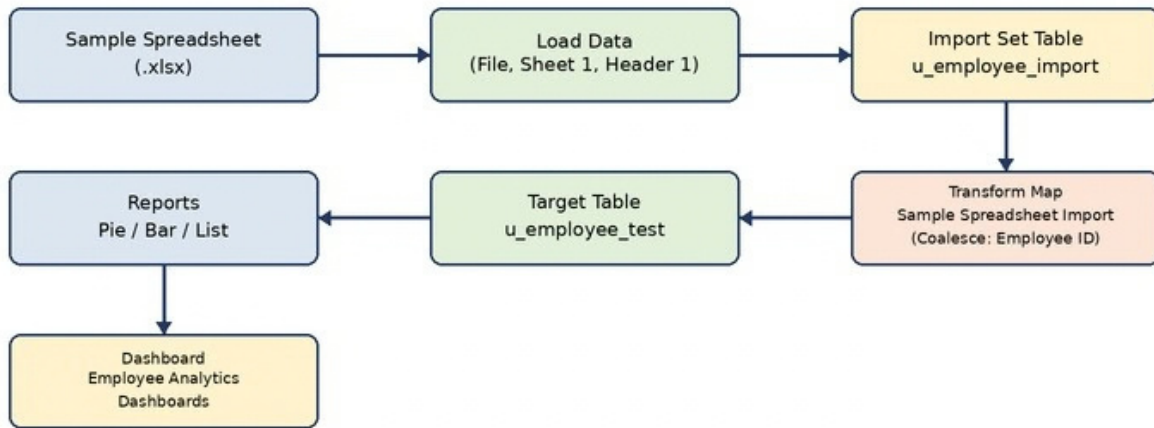


Figure 1: Data flow from spreadsheet to dashboard

3.4 Technology Stack

Component	Technology / Feature
Platform	ServiceNow
Data source	Google Sheets, exported as Excel (.xlsx)
Staging table	Import Set table: Employee Import (u_employee_import)
Target table	Custom table: Employee Test (u_employee_test)
Transformation	Transform Map: Sample Spreadsheet Import, Mapping Assist, Coalesce
Verification	Transform History, Employee Test list view
Analytics	Reports (Pie, Bar, List) and Dashboard (Employee Analytics Dashboards)

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem	Solution
Bulk employee data arrives as Excel files	Import Sets load the file into a staging table
Source columns differ from target fields	Transform Map with Auto Map and Mapping Assist links source to target fields
Repeated imports create duplicates	Coalesce on Employee ID updates existing records and ignores unchanged ones
No quick view of the data	Reports and a dashboard show department, location and list views

4.2 Proposed Solution

The proposed solution is a staged import pipeline built entirely with ServiceNow configuration:

- **Prepare:** create the sample spreadsheet in Google Sheets and download it as .xlsx.
- **Target table:** create Employee Test (u_employee_test) with Employee ID, Employee Name, Email, Department and Location.
- **Import Set:** use Load Data to create Employee Import (u_employee_import) and load the file.
- **Transform:** create the Transform Map "Sample Spreadsheet Import", map the fields and run the transform.
- **Validate:** confirm the records in Employee Test, then enable Coalesce and re-import to test inserts, updates and ignored rows.
- **Analyze:** build three reports and add them to the Employee Analytics Dashboards.

4.3 Solution Architecture

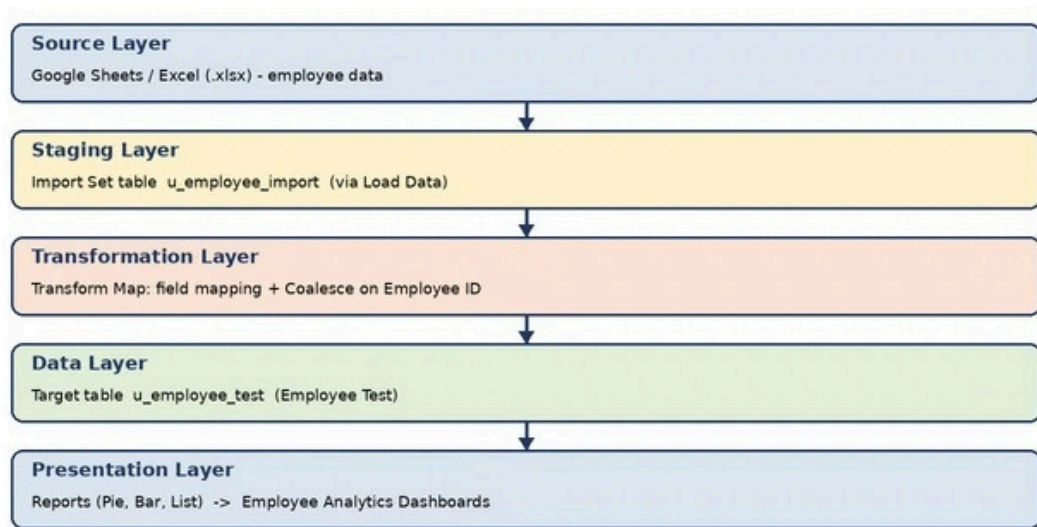


Figure 2: Layered solution architecture

The architecture separates the source file, the staging table, the transformation logic, the final data table and the presentation layer. This keeps raw data isolated from validated data and makes the import repeatable.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activity	Output
Phase 1	Prepare the sheets in Google Sheets; download as .xlsx	Sample Spreadsheet.xlsx
Phase 2	Create custom table Employee Test and its 5 fields	u_employee_test table
Phase 3	Load Data: create the import set table	u_employee_import table
Phase 4	Create Transform Map, map fields, run Transform	Success message

Phase 5	Transform data and validate in Employee Test	Records created in target table
Phase 6	Enable Coalesce on a field map	Duplicate prevention
Phase 7	Insert new/updated Excel data and re-run the transform	Inserted / updated / ignored counts
Phase 8	Create 3 reports on Employee Test	Department, Location and List reports
Phase 9	Create dashboard and add the reports	Employee Analytics Dashboards

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The import and transformation were tested with the following scenarios and checked in Transform History:

#	Test scenario	Observed result	Status
1	Initial import of Sample Spreadsheet.xlsx and Transform	State Completed and Success; all records created in Employee Test	Pass
2	Coalesce enabled; 4 rows uploaded (2 existing IDs with changed Email/Name, 2 new IDs)	2 rows inserted (new IDs), 2 rows updated (same IDs with changed values)	Pass
3	Same 4-row sheet uploaded again with the	Total 4 rows: 0 inserted, 0 updated, 4 ignored	Pass

	same Transform Map		
4	Reports and dashboard rendering	Pie, Bar and List reports display correctly on the dashboard	Pass

7. RESULTS

7.1 Output Screenshots

The screenshot shows a mobile application interface for a spreadsheet. At the top, there is a title bar with a close icon and the text 'DOC-20260929-WA0000'. Below the title bar is a toolbar with icons for save, share, and other functions. The main area displays a table with columns labeled 'employee_id', 'name', 'email', 'department', and 'location'. The table contains 12 rows of data. A green selection box highlights the cell in row 8, column 'H'. At the bottom, there is a formula bar with the text 'Enter text or formula here' and a search bar with the text 'Sheet1'.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	employee_id	name	email	department	location										
2	EMP001	John Doe	john.doe@IT	IT	New York										
3	EMP002	Alice Smith	alice.smith@HR	HR	London										
4	EMP003	Bob Jones	bob.jones@Finance	Finance	Tokyo										
5	EMP004	Charlie Wilson	charlie.wilson@IT	IT	Austin										
6	EMP005	Emily Miller	emily.miller@Sales	Sales	Sydney										
7	EMP006	Joseph Brown	joseph.brown@Marketing	Marketing	Austin										
8	EMP007	Mark Moore	mark.moore@HR	HR	Chicago										
9	EMP008	Barbara Davis	barbara.davis@IT	IT	New York										
10	EMP009	Barbara Rodriguez	barbara.rodriguez@Sales	Sales	Chicago										
11	EMP010	Nicole Brown	nicole.brown@Sales	Sales	Chicago										
12	EMP011	Rebecca Lee	rebecca.lee@Procurement	Procurement	Toronto										

Figure 3: Sample spreadsheet data (source)

The screenshot shows a database management interface. At the top, there is a header bar with the text 'Table Employee Test' and buttons for 'Delete', 'Update', and 'Delete All Records'. Below the header bar is a description of a table: 'A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More info](#)'. Below the description are fields for 'Label' (Employee Test), 'Name' (u_employee_test), 'Application' (Global), 'Remote Table' (checkbox), and 'In Data Interface' (checkbox). Below these fields is a section titled 'Columns' with a sub-section 'Table Columns for text'. This section contains a table with columns: 'Column label', 'Type', 'Reference', 'Max length', 'Default value', and 'Display'. The table lists three entries: 'Updated' (Date/Time, (empty), 40, false), 'Employee Name' (String, (empty), 40, false), and 'Email' (String, (empty), 40, false).

Column label	Type	Reference	Max length	Default value	Display
Updated	Date/Time	(empty)	40	false	false
Employee Name	String	(empty)	40	false	false
Email	String	(empty)	40	false	false

Figure 4: Employee Test table with created fields

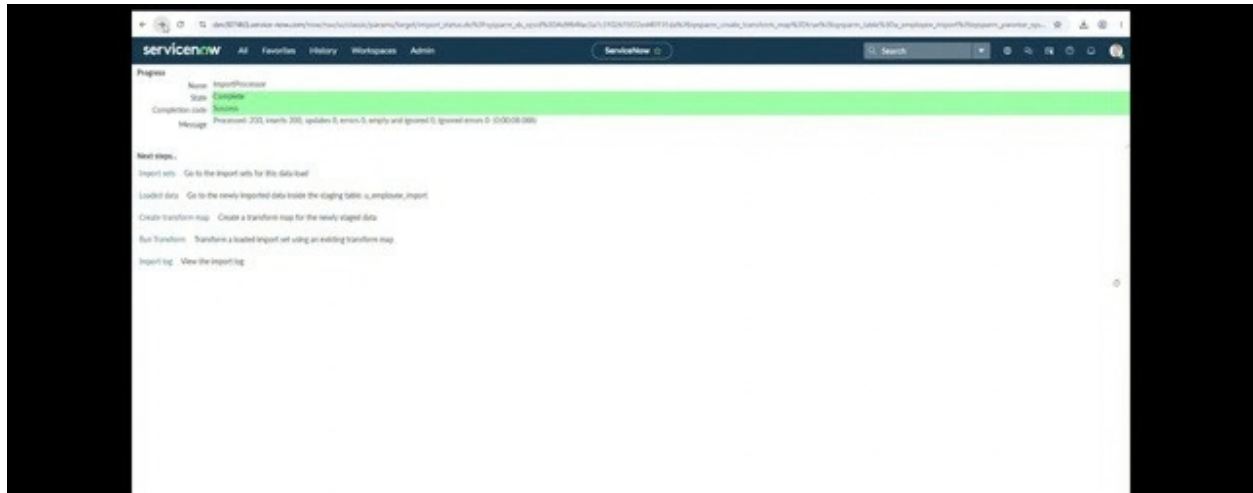


Figure 5: Import set table created successfully

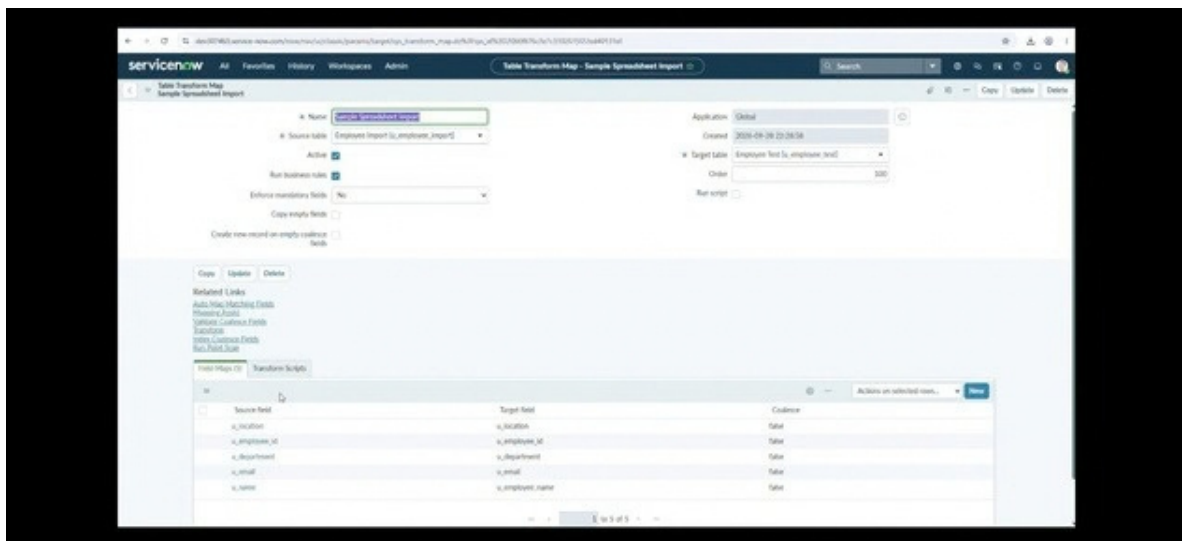


Figure 6: Transform Map "Sample Spreadsheet Import"

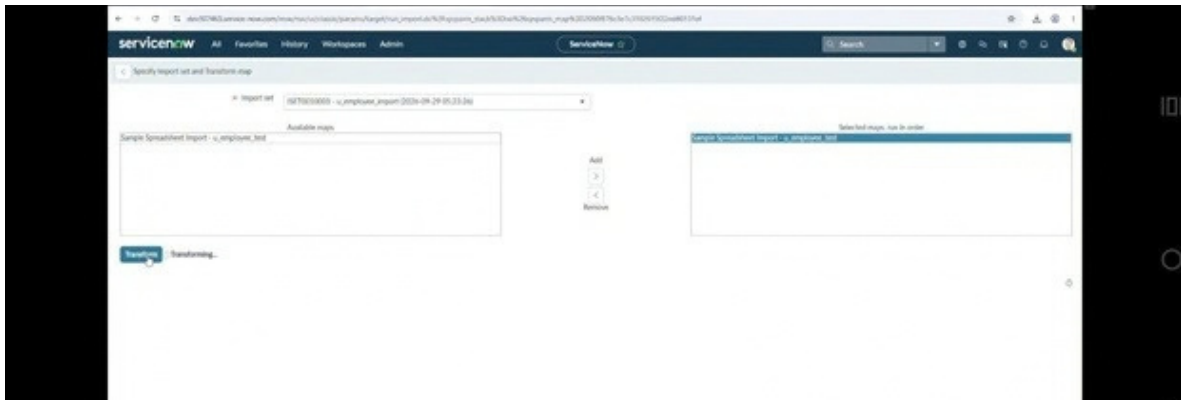


Figure 7: Mapping Assist (source to target fields)

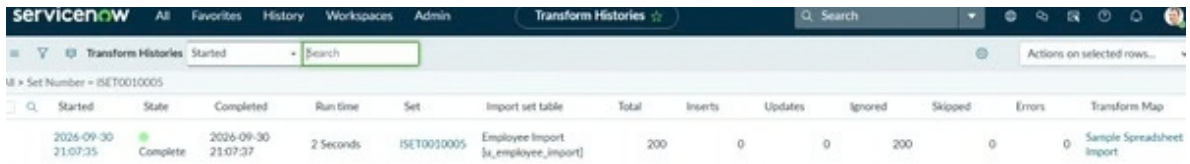


Figure 8: Transform completed successfully

The screenshot shows the ServiceNow Employee Table. The table has columns for Employee ID, Employee Name, Department, Email, and Location. The data is as follows:

Employee ID	Employee Name	Department	Email	Location
EMP004	Nicholas Baker	IT	nicholas.baker104@example.com	Berlin
EMP006	Joseph Wilson	Finance	joseph.wilson@example.com	Sydney
EMP023	Donald Rivers	Sales	donald.rivers23@example.com	Austin
EMP048	Betty Flores	Legal	betty.flores48@example.com	Paris
EMP082	Brian Rodriguez	Sales	brian.rodriguez82@example.com	Paris
EMP039	James Hill	Marketing	james.hill39@example.com	Austin
EMP029	Barbara Ramirez	Sales	barbara.ramirez29@example.com	Chicago
EMP044	Donald Hernandez	HR	donald.hernandez44@example.com	Singapore
EMP052	Mark Smith	HR	mark.smith52@example.com	New York
EMP021	Michelle Evans	Operations	michelle.evans21@example.com	Berlin
EMP005	Emily Miller	Sales	emily.miller@example.com	Sydney
EMP048	Sharon Wright	Operations	sharon.wright48@example.com	Austin
EMP027	Charles Smith	HR	charles.smith27@example.com	Berlin
EMP025	Elizabeth Torres	Sales	elizabeth.torres25@example.com	Sydney
EMP007	Jessica Carter	HR	jessica.carter07@example.com	Chicago
EMP006	Stephanie Baker	HR	stephanie.baker06@example.com	Austin
EMP021	Brandon Cook	Legal	brandon.cook21@example.com	Chicago
EMP023	Armando Scott	Finance	armando.scott23@example.com	Austin
EMP024	Cara Edwards	IT	cara.edwards24@example.com	London
EMP054	Pamela Stewart	Sales	pamela.stewart54@example.com	Singapore

Figure 9: Records imported into the Employee Test table



Started	State	Completed	Run time	Set	Import set table	Total	Inserts	Updates	Ignored	Skipped	Errors	Transform Map
2026-09-30 21:07:35	Complete	2026-09-30 21:07:37	2 Seconds	ISET0010005	Employee Import [u_employee_import]	200	0	0	200	0	0	Sample Spreadsheet Import

Figure 10: Transform History showing inserted, updated and ignored rows

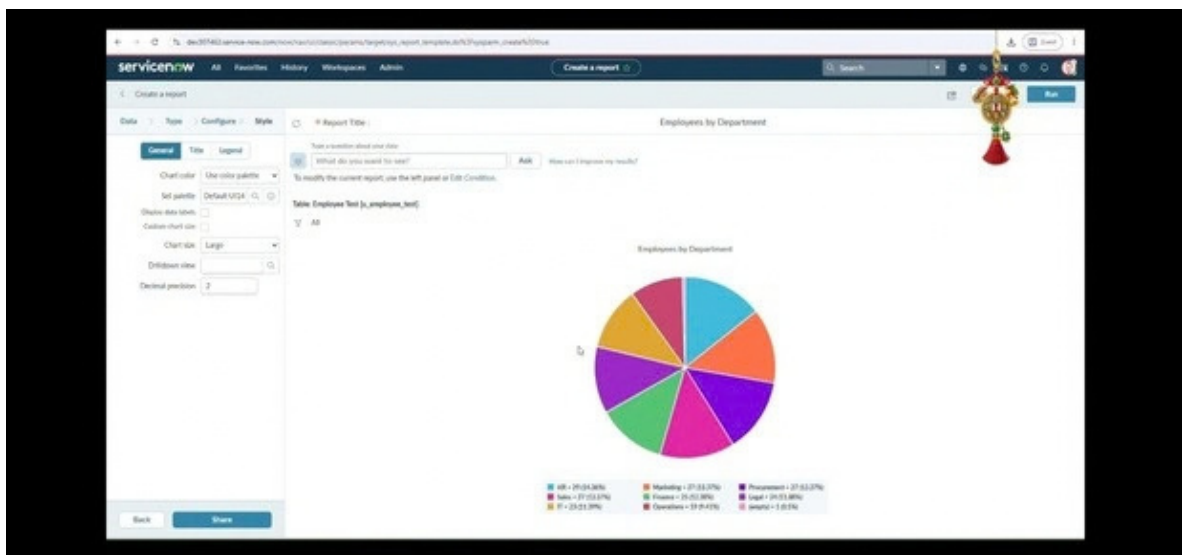


Figure 11: Report 1 - Employees by Department (Pie chart)

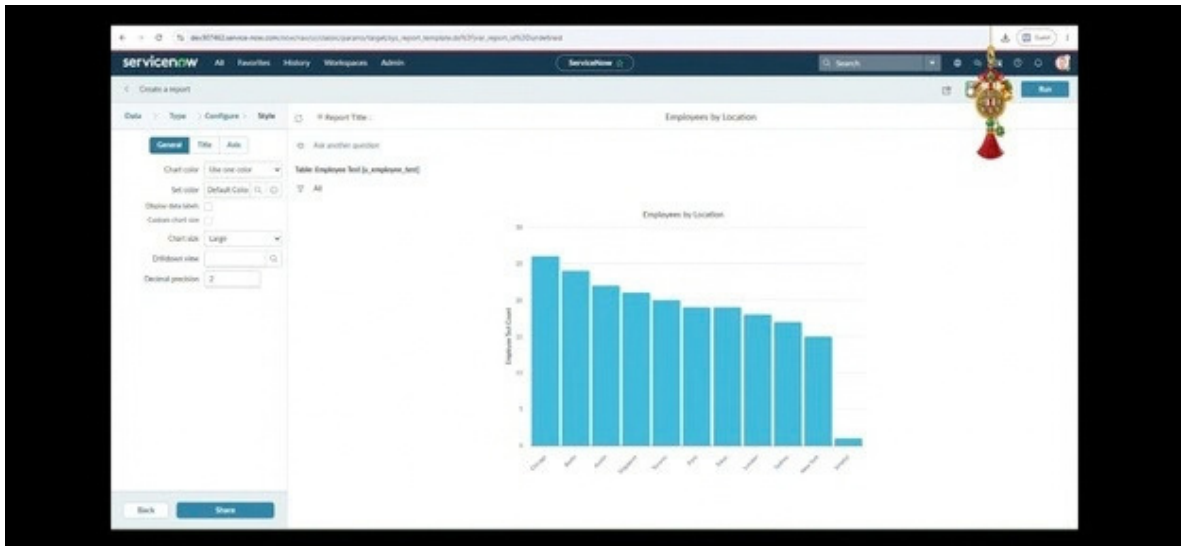


Figure 12: Report 2 - Employees by Location (Bar chart)

The screenshot shows the ServiceNow Report Designer interface. The report title is 'Employees list report'. The data source is 'Table: Employee Test [u_employee_test]'. The report type is 'List'. The report shows a table with 5 columns: Employee ID, Employee Name, Email, Department, and Location. The table contains 10 rows of employee data.

Employee ID	Employee Name	Email	Department	Location
EX49001	John Doe	john.doe@example.com	IT	New York
EX49002	Alice Smith	alice.smith@example.com	HR	London
EX49003	Bob Jones	bob.jones@example.com	Finance	Tokyo
EX49004	Charlie Williams	charlie.williams@example.com	IT	Austin
EX49005	Emily Miller	emily.miller@example.com	Sales	Sydney
EX49006	Joseph Brown	joseph.brown@example.com	Marketing	Austin
EX49007	Mark Moore	mark.moore@example.com	HR	Chicago
EX49008	Barbara Diaz	barbara.diaz@example.com	IT	New York
EX49009	Barbara Ramirez	barbara.ramirez@example.com	Sales	Chicago
EX49010	Nicole Brown	nicole.brown@example.com	Sales	Chicago
EX49011	Rebecca Lewis	rebecca.lewis@example.com	Procurement	Toronto
EX49012	Paula Smith	paula.smith@example.com	Finance	Berlin

Figure 13: Report 3 - Employee List Report

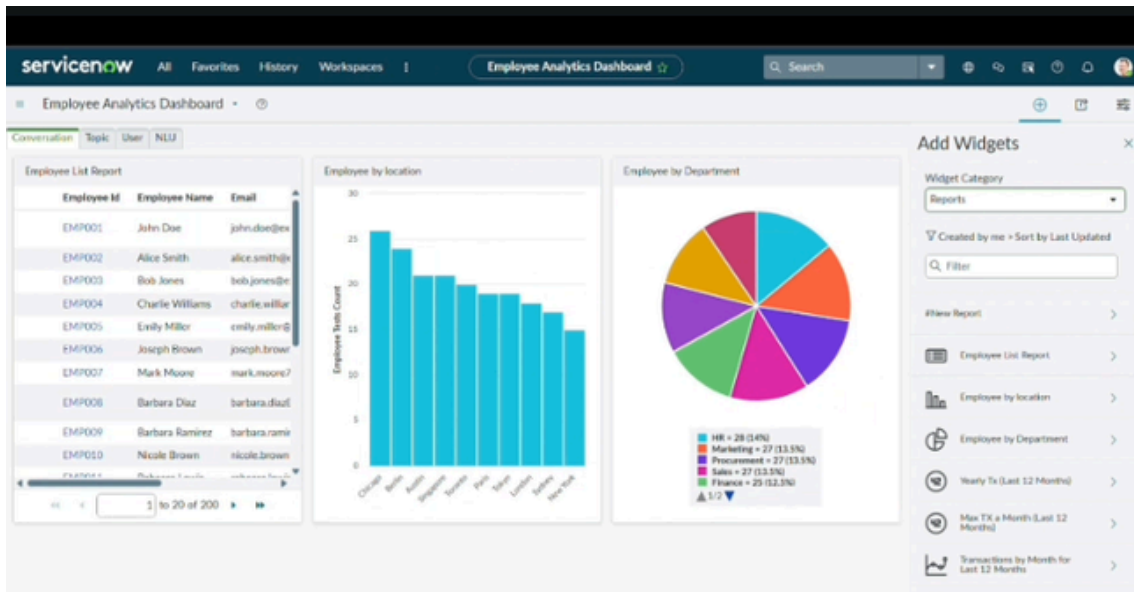


Figure 14: Employee Analytics Dashboards with all three reports

8. ADVANTAGES & DISADVANTAGES

Advantages

- Fast bulk loading of employee data without manual entry.
- Staging in an Import Set table protects the target table from raw, unvalidated data.
- Coalesce prevents duplicates and keeps existing records up to date.
- No coding needed; the process is easy for beginners to follow and repeat.
- Reports and dashboards give quick, visual insight into the data.

Disadvantages

- Requires manual upload of the file each time; the process is not automated.
- Source column headers and data formats must match the expected structure.
- All fields are String type, so there is no built-in format validation (for example, of email addresses).
- Creating dashboards via the backend form requires Admin Overrides to be enabled on the ACL.

9. CONCLUSION

This project successfully implemented an employee data management solution using ServiceNow Import Sets and Transform Maps, ensuring accuracy, efficiency and data integrity during employee onboarding and updates.

A key enhancement was the use of Coalesce in the Transform Map. By enabling Coalesce on the Employee ID field, the system identifies existing employee records

on every import, so repeated imports update existing data instead of inserting redundant entries. This keeps the employee database clean, reliable and consistent, which is critical in real-world environments where data may be imported multiple times.

The project was also extended with Reports and a Dashboard. The reports show employee distribution by department and location along with a full employee list, and they are consolidated into the Employee Analytics Dashboards. This lets administrators and HR teams monitor data trends and data quality from a single view and reduces the need for manual analysis.

10. FUTURE SCOPE

- Schedule automated imports (for example, scheduled data imports from a file server or SFTP) instead of manual uploads.
- Add field-level validation and transform scripts (for example, email format checks, standardizing department names).
- Report on import activity, such as total, newly added, updated and failed records, and add these to the dashboard.
- Map imported employees to the ServiceNow User table and add role-based access to the dashboard.
- Set up notifications for failed or partially successful imports.

11. APPENDIX

[Demo link](#)

https://drive.google.com/file/d/17T74PIKrGQca98RaXTd0qxJVEKa_tvLY/view?usp=drivesdk